Confusion Matrix-Simple Summary

➢ Random Forest

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.95	0.92	0.94	79
1	0.86	0.90	0.88	41
accuracy			0.92	120
macro avg	0.90	0.91	0.91	120
weighted avg	0.92	0.92	0.92	120

1. **Out of all predicted as purchased, how many actually purchased?**
   Precision: 86%
2. **Out of all actual purchased cases, how many were correctly identified?**
   Recall: 90%
3. **What percentage of all predictions are correct overall?**
   Accuracy: 92%
4. **Which metric represents the balance between correctness and completeness?**
   F1-score: 0.88
5. **How reliable are the model's predictions in general?**
   Precision: 86%
6. **How effectively are actual outcomes captured by the model?**
   Recall: 90%
7. **What is the average performance across all outcomes equally?**
   Macro Avg F1-score: 0.91
8. **What is the performance considering data imbalance?**
   Weighted Avg F1-score: 0.92
9. **Which metric reflects overall balanced performance?**
   F1-score: 0.88
10. **Which metric represents overall correctness of the model?**
    Accuracy: 92%

➢ Decision Tree

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.91	0.94	0.93	79
1	0.87	0.83	0.85	41
accuracy			0.90	120
macro avg	0.89	0.88	0.89	120
weighted avg	0.90	0.90	0.90	120

1. **Out of all predicted as purchased, how many actually purchased?**
   Precision: 87%
2. **Out of all actual purchased cases, how many were correctly identified?**
   Recall: 83%
3. **What percentage of all predictions are correct overall?**
   Accuracy: 90%
4. **Which metric represents the balance between correctness and completeness?**
   F1-score: 0.85
5. **How reliable are the model's predictions in general?**
   Precision: 87%
6. **How effectively are actual outcomes captured by the model?**
   Recall: 83%
7. **What is the average performance across all outcomes equally?**
   Macro Avg F1-score: 0.89
8. **What is the performance considering data imbalance?**
   Weighted Avg F1-score: 0.90
9. **Which metric reflects overall balanced performance?**
   F1-score: 0.89
10. **Which metric represents overall correctness of the model?**
    Accuracy: 90%

➢ Support Vector Machine

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.66	1.00	0.79	79
1	0.00	0.00	0.00	41
accuracy			0.66	120
macro avg	0.33	0.50	0.40	120
weighted avg	0.43	0.66	0.52	120

1. **Out of all predicted as purchased, how many actually purchased?**
   Precision: 0%
2. **Out of all actual purchased cases, how many were correctly identified?**
   Recall: 0%
3. **What percentage of all predictions are correct overall?**
   Accuracy: 66%
4. **Which metric represents the balance between correctness and completeness?**
   F1-score: 0.00
5. **How reliable are the model's predictions in general?**
   Precision: 66%

6. **How effectively are actual outcomes captured by the model?**
   Recall: 100%
7. **What is the average performance across all outcomes equally?**
   Macro Avg F1-score: 0.40
8. **What is the performance considering data imbalance?**
   Weighted Avg F1-score: 0.52
9. **Which metric reflects overall balanced performance?**
   F1-score: 0.40
10. **Which metric represents overall correctness of the model?**
    Accuracy: 66%